Introduction to ethical framework 3: Deontology Ethics and Machine Learning

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Disclosures


- Anil Raj, MD has no conflicts to report
- Univ of Michigan (BA,MD)
  - RA- Jenks Vestibular Lab, MEEI, Boston
  - RA- Vestibular Testing Lab, UofM Medical Center
- University of Hawai'i (General Surgery PGY-1)
- Tripler Army Medical Center (General Medical Officer)
- Johnson Space Center, Neurosciences Lab (NRC Resident Research Fellow)
- Naval Aerospace Medical Research Lab (Research Scientist)
- Research Scientist, IHMC

Locations & Affiliations
Pensacola
University of West Florida
UNF
UNIVERSITY of NORTH FLORIDA
UCF
FLORIDA INSTITUTE OF TECHNOLOGY
1958
USF
UNIVERSITY OF SOUTH FLORIDA
Ocala
FAU
FLORIDA ATLANTIC UNIVERSITY
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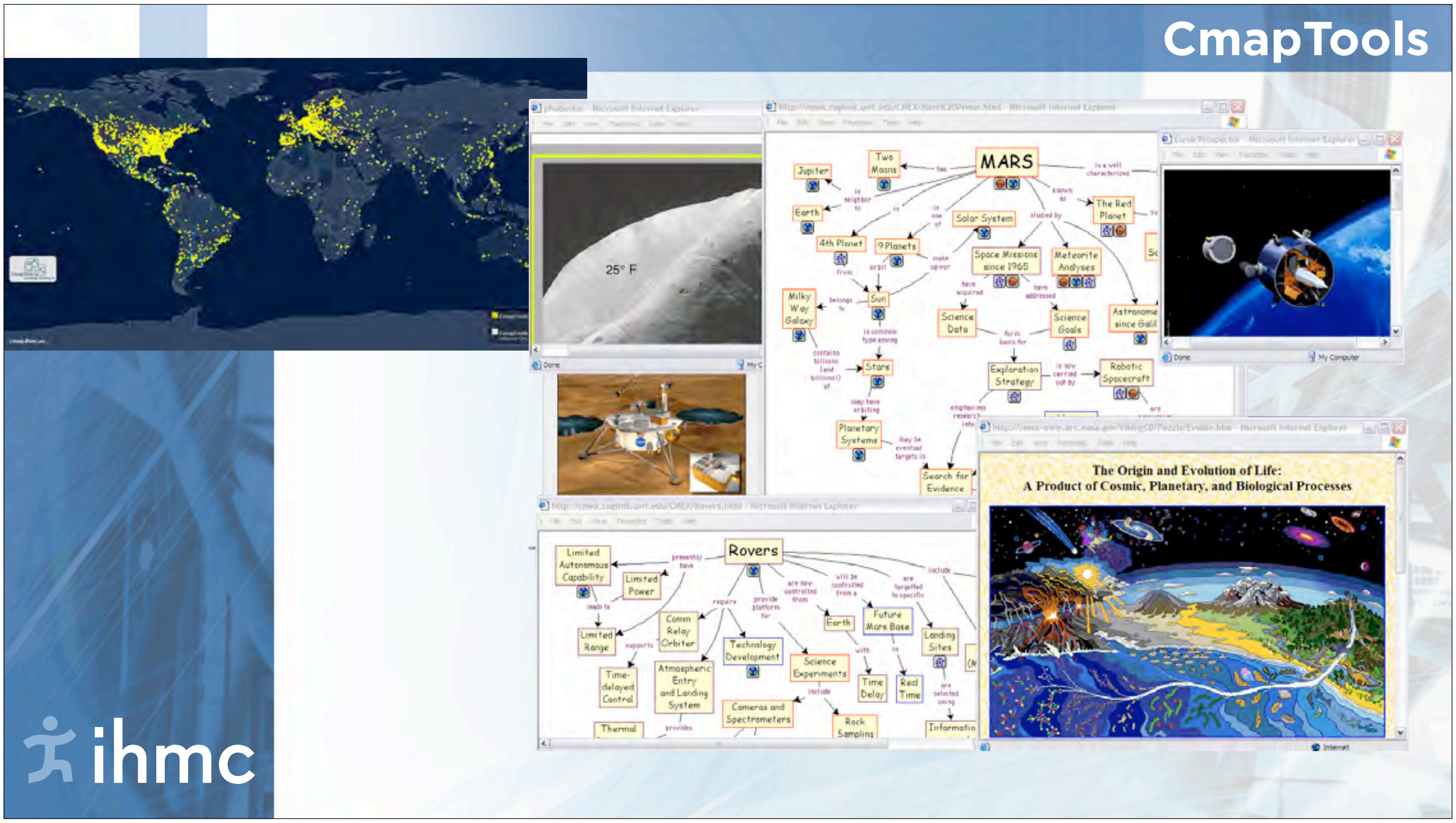
Human Centered Systems
Research that makes a difference...
that makes a difference


CmapTools
MARS
Rovers
The Origin and Evolution of Life:
A Product of Cosmic, Planetary, and Biological Processes
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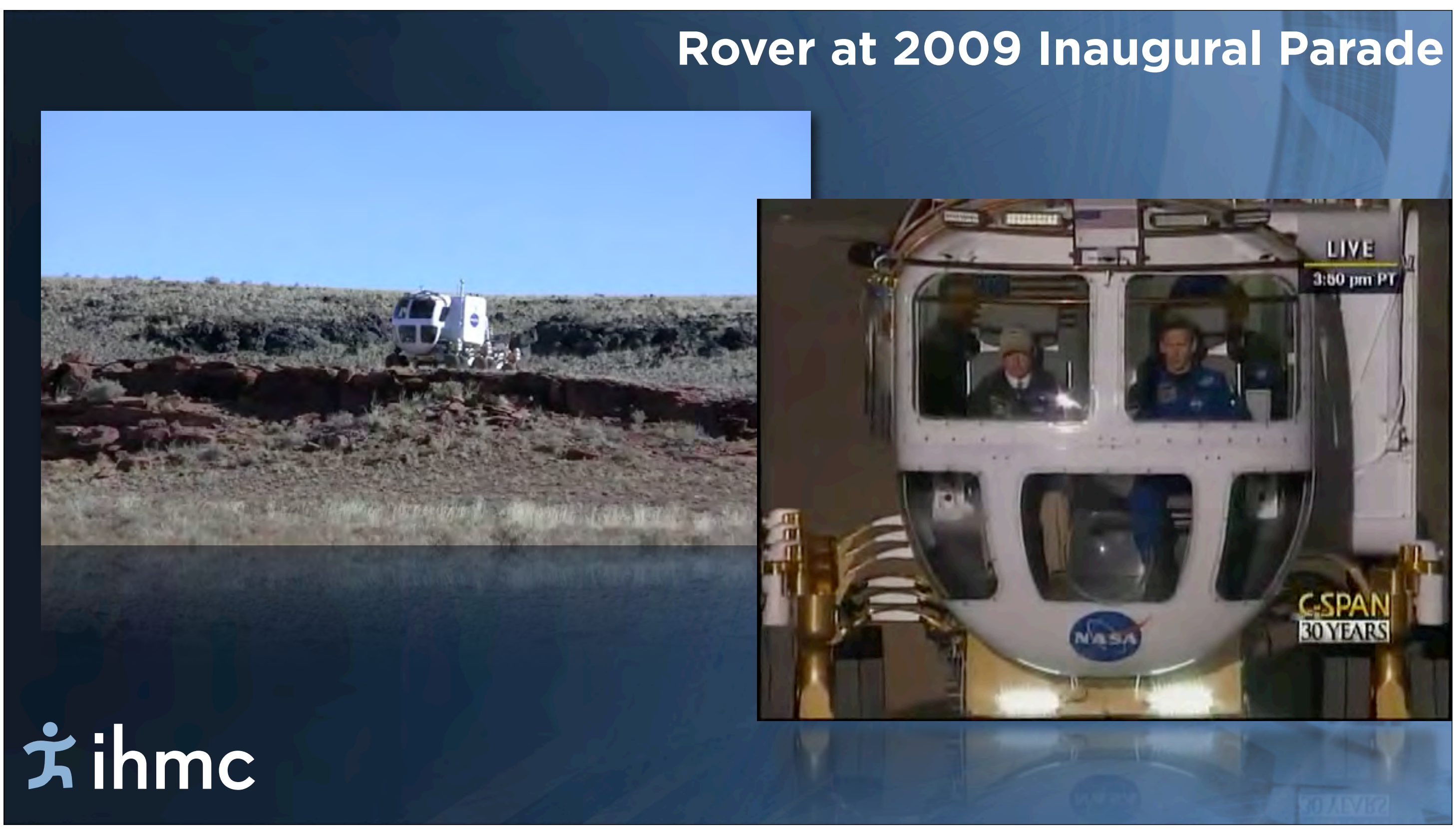
Rover at 2009 Inaugural Parade
LIVE
3:50 pm PT
C-SPAN
30 YEARS
NASA
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Cybathlon 2016

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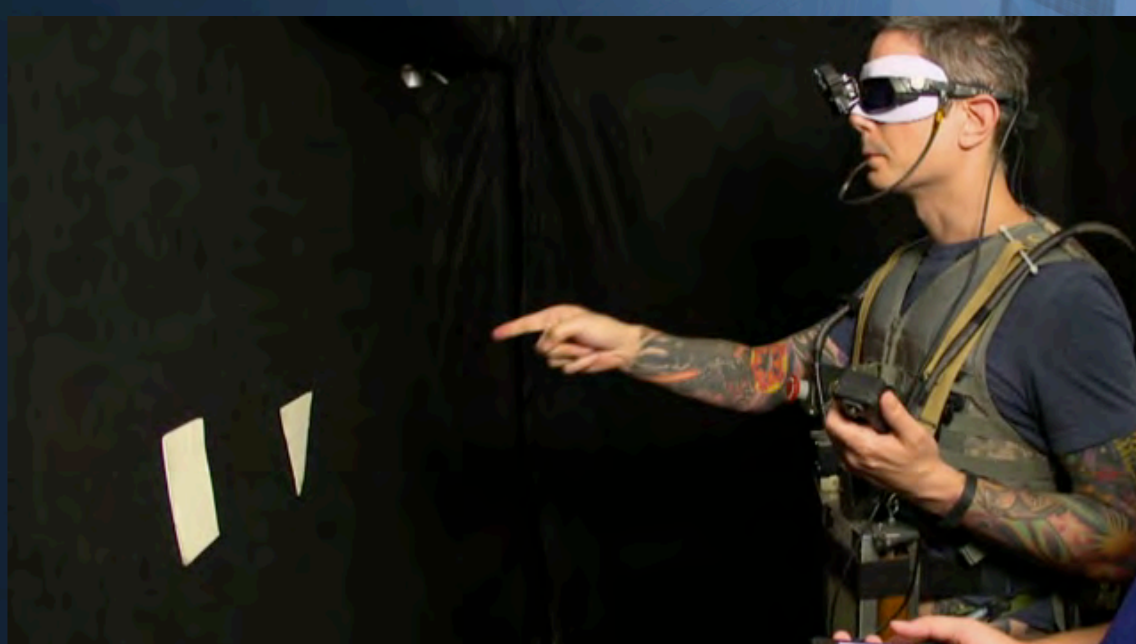
Multichannel Tactile Integration



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Terminology

- Deontology
  - The study (*logos*) of duty (*deon*)
- Machine Learning
  - A field of Artificial Intelligence that relies on statistical methods and algorithms to build inference models of tasks
  - Enables identification of feature within a data set without explicit instructions

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Deontology

- Focuses on determining the "right" course of action rather than focusing on the "goodness" of the outcome
  - The ends (outcomes) must be obtained through just means, even if they result in ends that are considered less good
- Contrasts with Consequentialism
  - Focuses on the goodness of the outcome

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Deontology in Medicine

- Physicians have a duty to their patients
  - *Primum non nocere*: First do no harm
  - Some choices cannot be justified by their effects—that no matter how morally good their consequences

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Case 1

- Patient has a progressive terminal disease
  - Cancer in unsophisticated elderly patient
  - Family wishes to keep diagnosis secret
  - Consequentialist?
  - Virtue?
  - Deontologist?

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Case 2

- Patient has a progressive terminal disease
  - Huntington’s Chorea in educated young patient
  - Genetic screening
  - Consequentialist?
  - Virtue?
  - Deontologist?

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Machine Learning in Medicine

- Genetic screening, DNA analysis, Bioinfomatics, etc., expanding with application of Machine Learning techniques
- How do we manage patient rights and privacy when using Machine Learning?

Machine Learning in Medicine

- Machine learning identification of medical conditions
  - Overt
    - Use of ML for the benefit of the patient/population
  - Covert
    - Infer medical information from public/private sources
- Total Information Awareness

Machine Learning in Medicine

- Career development
  - Change goals
- Family planning
  - Children
- Employment
  - Hire/retain employees
- Privacy/monetization of data
  - DNA registries/Doxing

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Privacy/monetization

- Who shares your data?
  - DNA registries used to identify individuals from DNA provided by relatives
  - Doxing from publicly available information
    - Apple, Google cellphone tracking
    - “Free” apps monetize personal data

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Mundane Machine Learning

- Natural language processing (NLP)
  - Siri, Alexa, Hey Google
    - Statistical, population based
    - Biased toward most common statements
  - Search engines
    - Search engine optimization
      - Uprank desired result by inferring the search algorithm
      - Twitter bots manipulating ranking

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Mundane Machine Learning

- Image processing
  - Facial recognition
    - Biased toward training data set
    - Who manages security of the biometrics
    - Social engineering
      - Crowd search without consent
    - Social media (Facebook, etc.)
      - Filters vs. freedom of expression

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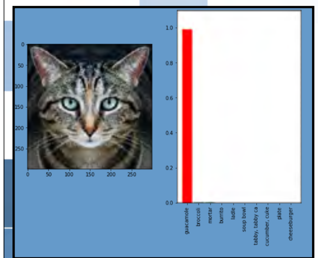
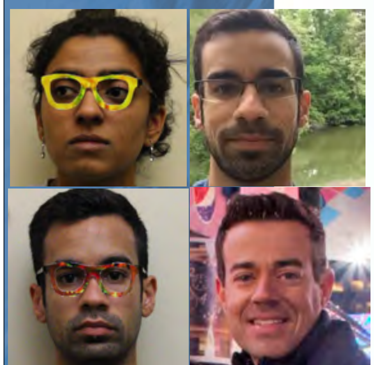
Deontology & Machine Learning





- Facial recognition in public places
  - Deontology challenge for goodness
    - Identify criminals?
      - Spoofing
      - Stale training data
      - Correlative versus causative features
    - Justice
      - AI driven judicial sentences

Deontology & Machine Learning

- Identify social infractions
  - Jaywalkers
  - Debtors
  - Litterers
  - Do the means justify these ends?
    - Medical conditions
    - Parking tickets
    - Poor grades at Uni

Deontology & Machine Learning

- Similar issues arise for medicine
  - Use medical ML for identification of outbreaks (e.g., influenza) before epidemic
  - Absolute security of data not practical
  - Re-identification of de-identified data enabled by ML
  - Identity theft, extortion

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Deontology & Machine Learning

- Classical philosophical thought experiments apply to ML and AI, but without moral agent
- Self-driving cars
  - Trolley problem
  - Transplant problem
- The programmer's implementation must enforce the ethics

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Deontology & Machine Learning

- Asimov's Three Laws of Robotics
  - A robot may not injure a human being or, through inaction, allow a human being to come to harm
  - A robot must obey the orders given it by human beings except where such orders would conflict with the First Law
  - A robot must protect its own existence as long as such protection does not conflict with the First or Second Laws

Deontology & Machine Learning

- Asimov's Three Laws of Robotics
  - Self driving cars should injure neither the human occupants nor individuals outside the car
  - Trolley problem
  - Who's responsible
    - Uber?
    - Pedestrian?
    - Coder?
    - Software?

Deontology & Machine Learning

- Problems extend beyond ethics
  - Legal vs. ethical responsibility
  - Corporate vs. individual
  - Financial
  - Social costs
    - Self-driving cars could increase safety and reduce traffic
    - Loss of trust will delay acceptance & more drivers/pedestrians will die

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Deontology & Machine Learning

- Data rights have intrinsic value
  - Population data exists as a national resource
  - Similar to minerals, petroleum, etc.
- Is the risk of exploitation/colonialism worth the benefits?

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Deontology & Machine Learning

- Costs of Machine Learning
  - Financial
    - CPU/GPU Investment
    - Facilities costs
  - Compute time
  - Carbon emissions

Consumption	CO_2e (lbs)
Air travel, 1 passenger, NY↔SF	1984
Human life, avg, 1 year	11,023
American life, avg, 1 year	36,156
Car, avg incl. fuel, 1 lifetime	126,000
Training one model (GPU)	
NLP pipeline (parsing, SRL)	39
w/ tuning & experimentation	78,468
Transformer (big)	192
w/ neural architecture search	626,155

Deontology & Machine Learning

- Ils doivent envisager qu'une grande responsabilité est la suite inséparable d'un grand pouvoir (1793)
  - ...great responsibility follows inseparably from great power
- 1962
- 2019?

Conclusions & Next Steps

- ML already has myriad use cases
  - Usage will expand as data processing, storage and manipulation capabilities increase
- ML has potential for providing great benefit to society
- ML can be easily exploited to the detriment of individuals or groups
- AI and ML must incorporate ethics

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